

格林公式

$$\oint_C P(x,y)dx + Q(x,y)dy = \iint_D \left(-\frac{\partial P}{\partial y} + \frac{\partial Q}{\partial x} \right) dx dy$$

$\partial D = C$

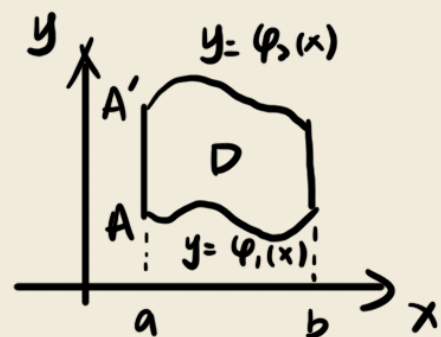
格林公式的证明

设有界闭区域 D 由逐段光滑曲线 C 围成, 函数 $P(x,y)$ $Q(x,y)$ 在 D 上具有一阶连续偏导数

$$\text{则 } \oint_C Pdx + Qdy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

$$\text{先证 } \oint_C Pdx = - \iint_D \frac{\partial P}{\partial y} dx dy$$

① 区域 D 由曲线 $y = \varphi_1(x)$, $y = \varphi_2(x)$ 及 $x=a$, $x=b$ 围成



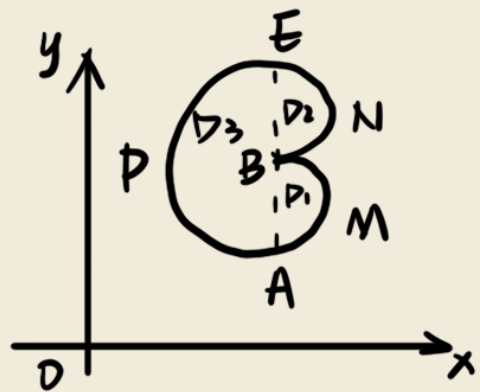
$$\begin{aligned} \oint_C Pdx &= \int_{\widehat{AB}} Pdx + \int_{\widehat{BB'}} Pdx + \int_{\widehat{B'A'}} Pdx + \int_{\widehat{A'A}} Pdx \\ &= \int_a^b P(x, \varphi_1(x)) dx + 0 + \int_b^a P(x, \varphi_2(x)) dx + 0 \\ &= - \int_a^b (P(x, \varphi_2(x)) - P(x, \varphi_1(x))) dx \end{aligned}$$

而 = 重积分

$$\iint_D \frac{\partial P}{\partial y} dx dy = \int_a^b dx \int_{\varphi_1(x)}^{\varphi_2(x)} \frac{\partial P}{\partial y} dy = \int_a^b (P(x, \varphi_2(x)) - P(x, \varphi_1(x))) dx$$

$$\text{得 } \oint_C Pdx = - \iint_D \frac{\partial P}{\partial y} dx dy$$

② D 是单连通区域



$$\begin{aligned}
 - \iint_D \frac{\partial P}{\partial y} dx dy &= - \sum_{i=1}^3 \iint_{D_i} \frac{\partial P}{\partial y} dx dy \\
 &= \oint_{\widehat{AMBA}} P dx + \oint_{\widehat{BNEB}} P dx + \oint_{\widehat{EPAE}} P dx \\
 &= \int_{\widehat{AMB}} P dx + \int_{\widehat{BA}} P dx + \int_{\widehat{BNE}} P dx + \int_{\widehat{EB}} P dx + \int_{\widehat{EPA}} P dx + \int_{\widehat{AE}} P dx \\
 &= \left(\int_{\widehat{AMB}} + \int_{\widehat{BNE}} + \int_{\widehat{EPA}} \right) P dx = \oint_C P dx
 \end{aligned}$$

③ D 是多连通闭区域

分多连通为多个单连通区域即可

例: 求椭圆 $x = a\cos\theta, y = b\sin\theta$ ($0 \leq \theta \leq 2\pi$) 所围图形的面积

$$\text{取 } P(x, y) = -y \quad Q(x, y) = x$$

$$\text{得 } \oint xdy - ydx = 2 \iint dx dy$$

$$A = \frac{1}{2} \oint_C xdy - ydx = \frac{1}{2} \int_0^{2\pi} (ab\cos^2\theta + ab\sin^2\theta) d\theta = ab\pi$$

例: 求曲线积分 $\oint_C \frac{-(x+y)dx + (x-y)dy}{x^2+y^2}$ C 是不通过坐标原点的简单闭曲线, 取逆时针方向

$$P(x, y) = \frac{-(x+y)}{x^2+y^2} \quad Q(x, y) = \frac{x-y}{x^2+y^2}$$

$P(x, y), Q(x, y)$ 在 $O(0,0)$ 处无定义, 当 $(x, y) \neq (0,0)$ 时,

$$\frac{\partial Q}{\partial x} = \frac{y^2 + 2xy - x^2}{(x^2+y^2)^2} = \frac{\partial P}{\partial y}$$

① 若 C 所围区域不包含坐标原点,

$$\begin{aligned} \text{原式} &= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy \\ &= \iint_D 0 dx dy = 0 \end{aligned}$$

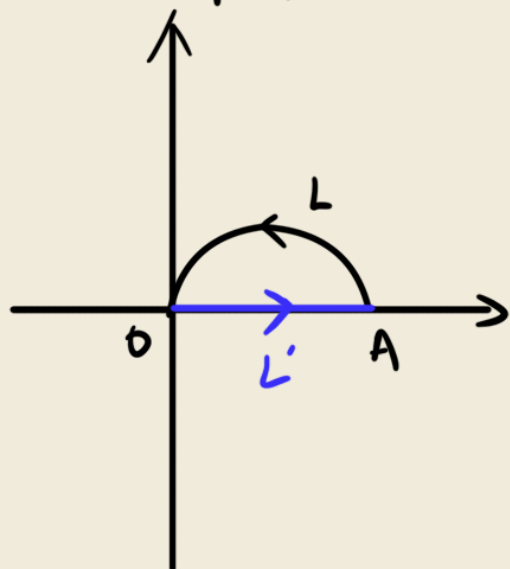
② 若 C 所围区域包含坐标原点

取 $r > 0$ 充分小, 使圆周 $l: x^2+y^2=r^2$ 位于区域内, l 取逆时针方向, 对 D_1 应用格林公式有

$$\int_{C+l} \frac{-(x+y)dx + (x-y)dy}{x^2+y^2} = \iint_{D_1} 0 dx dy = 0$$

$$\begin{aligned}
 &\Rightarrow \int_C \frac{-(x+y)dx + (x-y)dy}{x^2+y^2} = \int_L \frac{-(x+y)dx + (x-y)dy}{x^2+y^2} \\
 &= \int_0^{2\pi} \frac{-(r\cos\theta + r\sin\theta)(-r\sin\theta) + (r\cos\theta - r\sin\theta)(r\cos\theta)}{r^2} d\theta \\
 &= \int_0^{2\pi} d\theta = 2\pi
 \end{aligned}$$

例: 计算第=型曲线积分 $I = \int_L (e^x \sin y - 2y) dx + (e^x \cos y - 2) dy$, L 为 $x^2 + y^2 = ax$ ($a > 0$) 从 $(a, 0)$ 到 $(0, 0)$



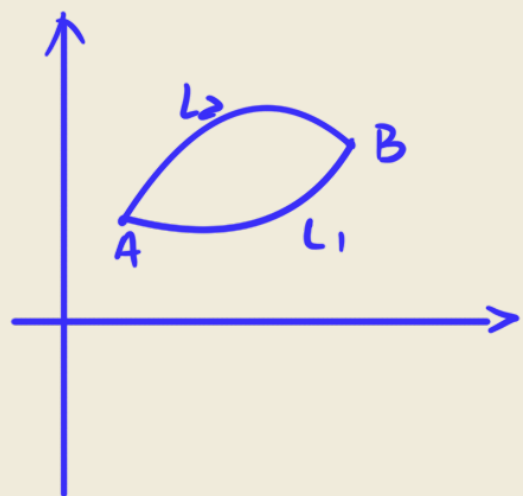
直接计算. 过于复杂 补一条 OA , 记为 L'

$$\oint_{L+L'} (e^x \sin y - 2y) dx + (e^x \cos y - 2) dy = \iint_D (e^x \cos y - (e^x \cos y - 2)) dx dy = 2 \iint_D dx dy = \frac{\pi}{4} a^2$$

$$= I + \int_0^a (e^x \cdot 0 - 2 \cdot 0) dx$$

$$\Rightarrow I = \frac{\pi}{4} a^2$$

平面上第二类曲线积分与路径无关的条件



$\int_C P(x,y) dx + Q(x,y) dy$ 的值与路径无关

$$\text{那 } \int_{L_1} P dx + Q dy = \int_{L_2} P dx + Q dy$$

设 D 是平面单连通区域, $P(x,y), Q(x,y)$ 在 D 内具有一阶连续偏导数, 则曲线积分 $\int_C P dx + Q dy$ 在 D 内与路径无关的充要条件是

$$\oint_L P dx + Q dy = 0$$

积分与路径无关 $\Rightarrow \oint_L P dx + Q dy = 0$ 充分不必要 无需区域单连通

设 D 是一单连通区域, 函数 $P(x,y), Q(x,y)$ 在 D 内具有一阶连续偏导数, 则 $P(x,y) dx + Q(x,y) dy$ 在 D 内恰是某一函数 $u(x,y)$ 的全微分的充要条件是 $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

必要性: $du = P(x,y) dx + Q(x,y) dy$

$$\text{则 } \frac{\partial u}{\partial x} = P(x,y), \frac{\partial u}{\partial y} = Q(x,y)$$

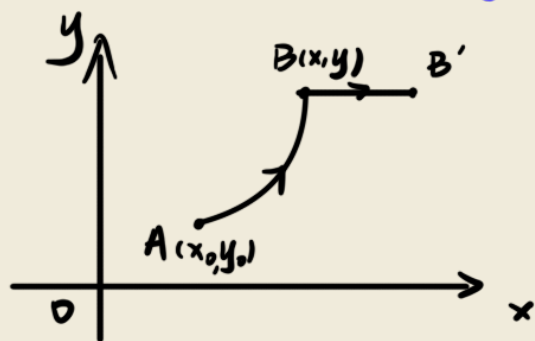
$$\text{从而 } \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial P}{\partial y}, \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial Q}{\partial x}$$

$$\text{一阶连续偏导数} \Rightarrow \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x} \Rightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

充分性: $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ 在 D 内恒成立 $\int_{AB} Pdx + Qdy$ 与路径无关

$$u(x, y) = \int_{(x_0, y_0)}^{(x, y)} P(x, y) dx + Q(x, y) dy$$

$$du = P(x, y) dx + Q(x, y) dy$$



$$u(x + \Delta x, y) = \int_{(x_0, y_0)}^{(x + \Delta x, y)} P dx + Q dy$$

$$u(x + \Delta x, y) - u(x, y) = \int_{(x, y)}^{(x + \Delta x, y)} P(x, y) dx + Q(x, y) dy$$

$$\text{积分中值定理} \Rightarrow u(x + \Delta x, y) - u(x, y) = P(\xi, y) \Delta x$$

D 为单连通区域, 函数 $P(x, y), Q(x, y)$ 在 D 内具有一阶连续偏导数, 对于 D 内任意两点 A, B , $\int_{AB} Pdx + Qdy$ 与路径无关的充要条件是 $Pdx + Qdy$ 恰为 $u(x, y)$ 的全微分

$$\int_{AB} Pdx + Qdy = \int_A^B du = u(B) - u(A)$$

例: 验证 $\frac{x dy - y dx}{x^2 + y^2}$ 在右半平面 $x > 0$ 内是某函数的全微分, 求出一个这样的函数

$$P(x, y) = \frac{-y}{x^2 + y^2} \quad Q(x, y) = \frac{x}{x^2 + y^2}$$

$$\frac{\partial Q}{\partial x} = \frac{y^2 - x^2}{(x^2 + y^2)^2} = \frac{\partial P}{\partial y}, \quad x^2 + y^2 \neq 0$$

为方便计算, 取从 $(1, 0)$ 到 (x, y) 右半平面. 不可取 $(0, 0)$

$$\text{取 } u(x, y) = \int_1^x \frac{-0}{t^2} dt + \int_0^y P \cdot 0 + \frac{x}{x^2 + t^2} dt = \int_0^y \frac{d(\frac{t}{x})}{1 + (\frac{t}{x})^2} = \arctan \frac{y}{x}$$

例: $P(x, y) = x^4 + 4xy^3$ $Q(x, y) = 6x^2y^2 + 5y^4$

(1) 验证在 xOy 平面内 $Pdx + Qdy$ 是某函数全微分

(2) 求 $Pdx + Qdy$ 原函数 $u(x, y)$

$$\text{1) } \frac{\partial P}{\partial y} = 12xy^2 = \frac{\partial Q}{\partial x} \Rightarrow Pdx + Qdy = du$$

$$\begin{aligned} \text{(2) } u(x, y) &= \int_{(0,0)}^{(x,y)} Pdx + Qdy = \int_0^x (t^4 + 0) dt + \int_0^y (6x^2t^2 + 5t^4) dt \\ &= \frac{1}{5}x^5 + 2x^2y^3 + y^5 \end{aligned}$$